

T&M CALIBRATION: Trends And Upgrades

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Technologies like automation and cloud connectivity are evolving, and so is the complexity of solutions

Test and measurement (T&M) devices can deliver accurate results as long as these are calibrated regularly. Precision is of utmost importance for any industrial test application. The performance of a complete system depends on the data represented by test devices. As T&M equipment in all industrial spheres evolve, calibration tools and technologies are upgrading simultaneously.

The importance of calibration

Appropriate calibration ensures test equipment's reliability by eliminating factors that may result in inaccuracy. This is done with the help of equipment called 'calibrators.' Test samples are measured using the device to be calibrated and the measured values are compared against the values known from the calibrator. Based on the difference in test results, adjustments are made to the test equipment.

Kalidas Bhangare, managing director of Testo India, elaborates, "Calibration of T&M equipment majorly involves parameters such as dew point, humidity, temperature and differential pressure. Some of the devices used at Testo's in-house calibration lab are Huminator for humidity calibration, Pneumator for pressure calibration, Drive Lock for high temperature measurement and Black Body for thermal imager calibration."

Fig. 1: Testo India's Huminator



Madhukar Tripathi, senior manager, Anritsu India, adds, "Calibration is an important execution on test equipment that confirms its current working performance and also ensures its long life. It is compulsory for anyone who values quality. Devices commonly used for calibration include power meters, power sensors, spectrum analysers and signal generators."

How device calibration is evolving

There is a lot of focus on reducing the time spent in T&M while maintaining precision. As a result, complexity of back-end design is on the rise.

Ranjit Nambiar, regional services manager, National Instruments, explains, "The T&M industry constantly demands capabilities like more accurate measurements and higher test capacities. R&D labs of T&M solution providers like us are working hard to launch products with specifications and features that match the feedback of the industry. As a result, we constantly provide T&M instruments that are more accurate and faster with increasing bandwidths."

Some of the major advancements in T&M calibration are listed below.

Increased frequency coverage. Calibration devices have the lowest and highest frequency thresholds within which these can work. For instance, Keysight Technology's two-port electronic calibration kit covers a frequency range of 300kHz to 9GHz, while the four-port setup works up to 20GHz.

"New technologies like 5G require higher frequency ranges and calibration instruments are upgrading accordingly. For instance, National Instruments has released vector signal analysers that cover frequency range up to 26.5GHz," informs Nambiar.

LAN taking up the GPIB space. Hitherto general-purpose interface bus (GPIB) had been the main mode of connecting calibration devices to computer systems. The devices used to have a GPIB port. The setup required a GPIB controller, which acted as an interface between the device and the computer, and a cable to connect to the computer.